

Verifone: Building Trust in the Second-Hand Phone Market

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ABSTRACT

With the rapid global growth of electronic waste, the potential for sustainable consumption in second-hand mobile phone markets remains largely untapped. This is due to a lack of trust caused by information asymmetry: potential buyers struggle to assess the quality of a used phone. To remedy this, we designed Verifone, a third-party interface that offers device-specific information for second-hand phones. With 3D interactions emulating tangible experiences and data visualisations aiming to increase transparency of the phone condition, Verifone instills confidence in the buying process. Our research and design process focused on understanding what elements of functional and affective design elicited more trust in the phone's condition. User evaluations proved that a clean, standardized design combined with affective elements to highlight device uniqueness could inspire more confidence in assessing the device's condition. The positive response suggests a tool like Verifone could encourage sustainable purchases, contributing to a circular economy. In addition, our research also sheds light on effective information presentation styles that build trust within the context of second-hand electronic purchases.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction design*.

KEYWORDS

sustainable consumption, second-hand markets, trust models, information asymmetry

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1 INTRODUCTION

The rise of technology brings with it an escalating issue of electronic waste. 61 million metric tons of electronic waste (e-waste) is projected to be produced this year [7], including 5.3 billion mobile phones [17], and is expected to increase further [7]. As the EU's Waste from Electrical and Electronic Equipment (WEEE) collection targets are weight-based [13] and recycling costs for smaller devices increase as their designs become more complex [14], governments are less motivated to invest in waste management initiatives for smaller devices compared to those of larger electronic devices. This could explain the low return and recycling rates for phones, estimated at 12-15% across most EU states [14]. This challenges the advancement of the UN's Sustainable Development Goal 12, centred on Responsible Consumption and Production [20].

Improved designs and manufacturer repairability programs [3, 19] enhance mobile phones' reuse and repair potential. The most recent EU commission recommendations emphasise the principle of driving a secondary market for small consumer electronics through financial incentives [12]. It also delineates tools that could assist consumers in understanding the value of used devices, such as a product passport [12]. However, consumer behaviour hinders reuse initiatives; only 17% of UK smartphones bought last year were second-hand [17], showing the market's potential is not fully realised.

Our primary and secondary research reveals a willingness to purchase second-hand phones and identifies various obstacles preventing this [16], chief among them a lack of trust in the condition of the phone and subsequent perceived risk. This is likely caused by information asymmetry, as buyers cannot access information about the device that sellers have. Most existing literature focuses on strategies to enhance consumer appeal of used phones through persuasion tactics, education initiatives and marketing efforts [15, 16]. Meanwhile, our research aims to address the issue of trust by balancing out information asymmetry.

Based on an iterative design process, we introduce Verifone, a third-party interface that offers comprehensive information about a specific used phone in an interactive format to bridge this information gap. Here we explore two presentation styles to help build trust: a functional design and an affective design.

2 RELATED WORK

2.1 The state of the second-hand phone market

There is ample supply of second-hand phones. On average, 2 abandoned phones are stockpiled in each UK household [16]. This is

largely attributed to their frequent and often premature replacement [7]. At the same time, purchasing demand for these used phones is low [17], directly impacting the growth of a circular economy [22]. This highlights an opportunity to promote the demand for used phones to extend their lifetime.

An in-depth survey conducted by Hubbub in October 2023 revealed that 64% of UK young adults recently bought a brand-new phone, often influenced by factors like extensive advertising or the indulgence in the unboxing experience [16]. Meanwhile, second-hand phones are considered a high-risk purchase due to the possibility of buying a malfunctioning phone. This same cohort did express a willingness to buy a used phone if these were more visibly promoted, offered higher cost savings, and were guaranteed to be in good condition. The appearance, durability, and battery life of second-hand phones were considered of utmost importance, alongside warranty expiry, the duration of software updates, and data cleansing. Notably, manufacturers could significantly contribute to the growth of the circular economy for second-hand phones. For example, Apple's refurbishment programme ensures genuine replacement parts, thorough cleansing and inspection, and a one-year guarantee. However, the price remains steep compared to other second-hand device seller [4].

2.2 Information Asymmetry & Trust

The second-hand phone market features information asymmetry: sellers know the device's condition, history, and potential issues – information buyers don't typically have easy access to. Notably, recent research shows that consumers are more likely to purchase a used phone if they receive more information about its prior use [16], reducing the perceived risk of the purchase.

In George Akerlof's influential work, "The Market for Lemons," he highlights how information gaps hinder fair trade and cause market failure [2]. In the used car market, buyers are cognisant of the existence of subpar cars ("lemons") and struggle to differentiate them from higher-quality cars ("plums"). Due to this lack of information, buyers become hesitant to pay higher prices, which in turn discourages the sellers of good cars from participating in the market. The paper advocates different remedies such as guarantees, branding and government regulation to mitigate this information-based uncertainty. This underscores the pivotal role of trust in these markets; in the absence of trust-building measures, they are prone to suffer.

In summary, the related work highlights: (1) Despite the abundant supply of used phones, there is insufficient demand, preventing responsible consumption. (2) Consumers are open to buying used phones but are dissuaded due to a lack of information on phone condition. (3) Measures to build trust can be taken to boost demand to stimulate a circular economy.

3 USER REQUIREMENTS

3.1 Exploratory User Research

To identify user attitudes towards e-waste, we conducted an exploratory survey of 52 UK residents aged 18-54. Cost was found to be the primary consideration when purchasing a phone. This highlights an opportunity to address e-waste by promoting the purchase of used phones, which often offer a similar feature set at

a lower cost than brand-new devices. Moreover, 58% of participants store old devices at home. These devices are excluded from the reuse and recycling process.

3.2 Problem Statement Generation

Initially, we focused on opportunities within the general e-waste context to pinpoint the most impactful area. We used an affinity diagram to synthesise our insights from both exploratory research and related works and identified potential problem statements. Plotting these statements on a 2x2 matrix (Figure 1 (a)), we chose the one with the highest impact on both the consumer and the environment. The statement selected was "I want to buy a second-hand phone because it is cost-efficient, but I do not trust the process" for the following reasons:

- High environmental impact: The manufacturing of new devices accounts for 85% of the carbon footprint of e-waste [8]. If the second-hand market becomes mainstream, this could prolong the lifespan of current devices, diminish landfill and toxic waste, and promote a circular economy.
- High consumer impact: The perceived unreliability of used phones deters interested consumers from purchasing devices at more reasonable costs. By addressing this issue, we can unlock a market that allows for substantial cost savings compared to buying new.

3.3 Defining User Needs

Our target audience is UK residents aged 18-34, as this cohort was identified to be more open to purchasing used phones [16]. To understand buyer pain points in the context of our problem statement, we defined a user persona (Figure 1 (b)).

Using this persona, we identified 3 user requirements to address the core need of trust: (1) Accessible: I want to easily access this information as part of the purchasing journey; (2) Reliable: I want to feel that the information presented to me is unbiased and accurate; (3) Clear: I want to fully understand the condition of the specific phone I am purchasing.

4 DESIGN PROCESS

Based on the user requirements, we conducted an iterative ideation process to explore ways of garnering trust in the process of purchasing second-hand phones.

4.1 Competitive Analysis

To guide our ideation, we first explored existing solutions in other second-hand markets:

CarFax is a third-party data aggregator providing transparency through a detailed PDF report of a used car's history based on its unique identifier, built to address information asymmetry in the used car market [10]. This encouraged us to consider a third-party certification to elicit trust. However, while a static PDF allows for a more standardised view of information, it lacks engaging interaction.

3DShot App by *Cappasity* is used in some second-hand fashion marketplaces to capture interactive 3D visuals of products, allowing buyers to assess their quality [9]. This offers a more tangible interaction, comparable to the in-store experience, but is unable

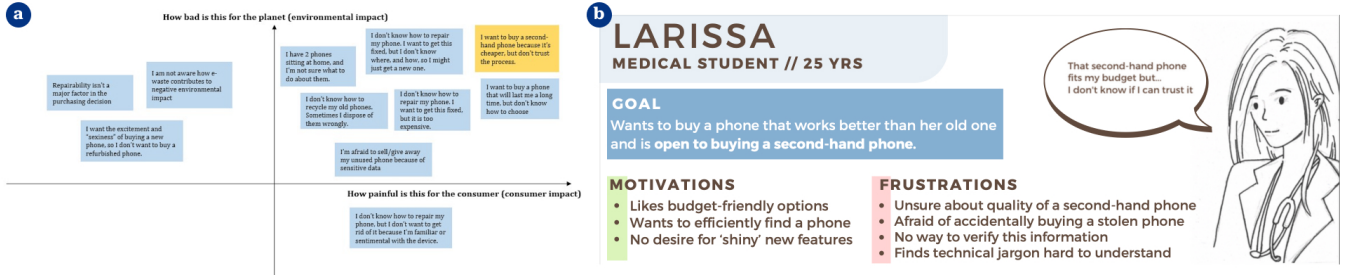


Figure 1: (a) 2x2 matrix for problems statements; (b) User persona – Larissa.

to communicate the history of prior use or internal condition of a device.

Vestiaire Collective is a second-hand designer fashion platform that manually vets the sellers and items for sale [11]. This offers credibility for products purchased on the platform, but the process is high-touch and not scalable. It adds significant friction for sellers, potentially discouraging them from entering the market altogether.

The exploration of solutions in comparable contexts enabled us to pinpoint interaction elements that could be applied to our problem space, as well as areas that needed deeper consideration. We determined our solution should: (1) provide device-specific information on condition and repair history; (2) have interactive elements for users to explore the specific phone's condition; (3) offer a scalable, frictionless solution for buyers within the existing market framework.

4.2 User Flow

Before delving into specific designs, we established how our solution would fit into the typical flow of purchasing used phones to meet our user requirement of accessibility. The solution would integrate within a marketplace or peer-to-peer platform with multiple access points: widgets or links on product listings, as well as a standalone platform. While concentrating on the buyer user flow, we discussed other stakeholders—sellers, repair shops, and manufacturers—who contribute the presented information. We envision a shared system for uploading information, enabling involved parties to add and modify details about the phone's health and repair history. This should be expanded on in future work.

4.3 Initial Ideas

We then conducted sketching activities such as Crazy 8s to explore different mediums for sharing device-specific information (Figure 2). This produced varied ideas, from an interactive dashboard and chatbot to Virtual Reality and Augmented Reality experiences. We then evaluated each idea and narrowed down the list according to their ability to fit our user requirements (Table 1).

The interactive dashboard scored highest in our evaluation, as its accessibility and clear manner of presenting information can elicit trust without much friction in the purchasing journey. However, it lacked the impact of a tangible experience which customers have with a model phone in store. Hence 3D interactive features were incorporated to communicate tangible aspects of the phone such as location of scratches and camera and audio quality.

Table 1: Evaluation criteria (scale of 1-10)

	Interactive Dashboard	Chatbot	Augmented Reality	Virtual Reality
Accessibility	10	7	2	1
Reliability	10	8	5	5
Clarity	8	3	8	8
Total	28	18	15	14

4.4 Information Delivery Style

Two distinct design styles emerged during the ideation process. One design included an anthropomorphised phone, meant to elicit an emotional response (affective), whilst the other adhered to familiar design patterns of popular phone manufacturers (functional). This sparked a discussion on trust-building elements, leading to additional literature review.

4.4.1 Functional Design & Implications. A functional design adheres to established standards and user expectations. It relies heavily on reflective design practices and prioritises efficiency, predictability and usability [21]. Our functional design aesthetic mimics that of Apple's website to evoke familiarity and trust; as research shows that consumers have high trust associated with well-known brands [18].

4.4.2 Affective Design & Implications. In affective design, the focus is on establishing an instant emotional bond with users. Our approach draws from visceral design [21], anthropomorphising the phone and framing it as an 'orphan' in need of a new home, inspired by Carfax's mascot redesign in 2010 [1]. Recognising the audience's positive response to a fox puppet in an ad campaign, Carfax became "a puppet with a purpose," [1] instilling confidence and advocating for fair value in customer purchases. However, there is an ongoing discussion on whether affective features effectively promote trust [23, 24].

Existing research indicates that both solutions could build trust, prompting our evaluation to identify which would be more effective in building trust within the context of the second-hand phone purchasing journey.

5 HIGH FIDELITY PROTOTYPES

To assess both designs, we created two high-fidelity interactive prototypes in Figma with identical structures and interactions but

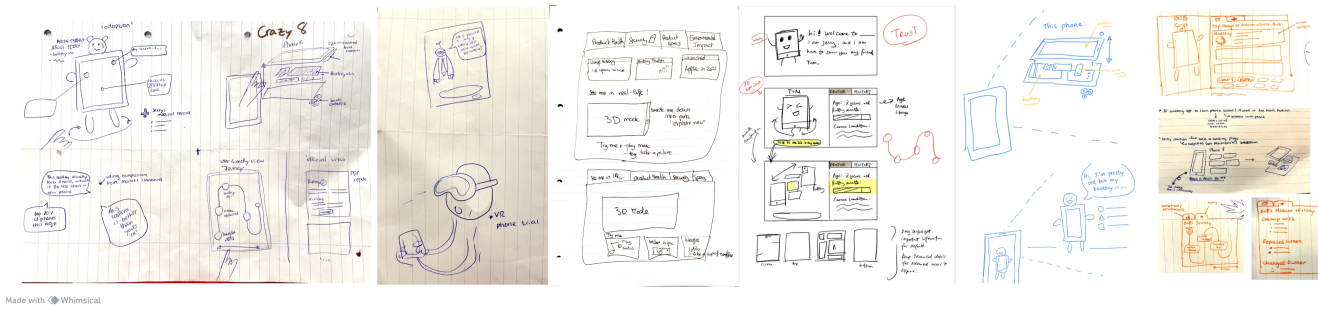


Figure 2: Crazy 8s and ideation sketches

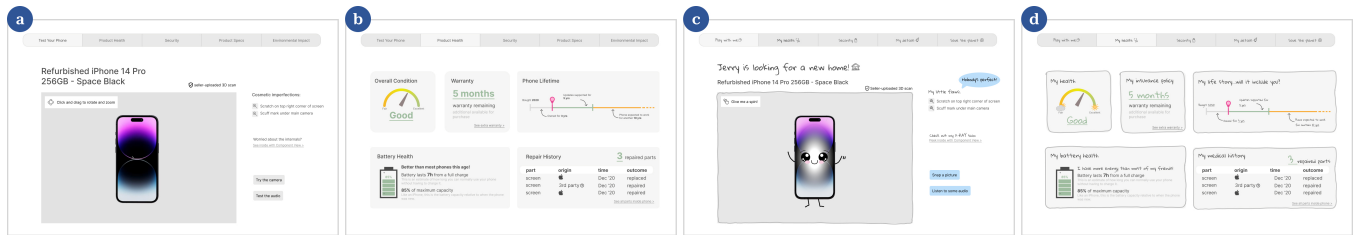


Figure 3: Comparisons between the two design styles. (a) Functional – landing page; (b) Functional – product health page; (c) Affective – landing page; (d) Affective – product health page.

different UX copy and visual elements, adhering to either functional or affective design. These prototypes fully demonstrate interactions on the landing and product health pages. We focused on developing these two pages as this would suffice to assess user preferences regarding perceived trust.

The functional prototype boasts a minimalist design with a monochromatic colour palette. It features a standard and recognisable typeface with concise copy (Figure 3). By contrast, in the affective prototype, we introduced more colours and emoticons, and a distinctive hand-written style typeface. Each phone was personified with a name, and its 3D model incorporated anthropomorphic elements such as a smiling face and waving arms. Playful copy was created for the titles and explanations.

6 EVALUATION & ITERATION

6.1 Usability Testing

We conducted usability tests with 6 participants recruited from our user survey to (1) assess Verifone’s trust-building effectiveness compared to a typical second-hand platform and (2) identify the preferred delivery style. The test consisted of 3 think-aloud segments, where participants were guided through a scenario of buying a second-hand phone. First, they navigated to a BackMarket product listing, a popular second-hand device marketplace [5]. Next, they explored the functional and affective prototypes freely in randomised order. Following each walkthrough, participants rated their confidence in information accuracy on a 1 to 7 scale, serving as a benchmark for measuring trust.

How confident are you that the information accurately reflects the condition of the phone?



Figure 4: Mean perceived trustworthiness scores for each prototype and the current state

6.2 User Feedback & Iteration

Participants expressed greater confidence in the accuracy of phone condition information in both prototypes, surpassing solely relying on the BackMarket product listing (Figure 4). As opposed to generic model information found on typical reselling platforms, Verifone offered device-specific details, which garnered trust and reassurance. Users appreciated the tangible experience and being able to visually assess cosmetic imperfections. Furthermore, the detailed breakdown of overall condition into sections (e.g., battery health) ensured data transparency.

The functional prototype was perceived as more trustworthy; participants highlighted its readability and clean aesthetics. The affective prototype generated mixed responses. While perceived as approachable, most participants expressed concern about the legitimacy of the information presented. Some found the playful elements distracting and hard to read. The handwritten elements appeared incongruent with the seriousness of the context. Nonetheless, the affective design did impart a unique personality to the

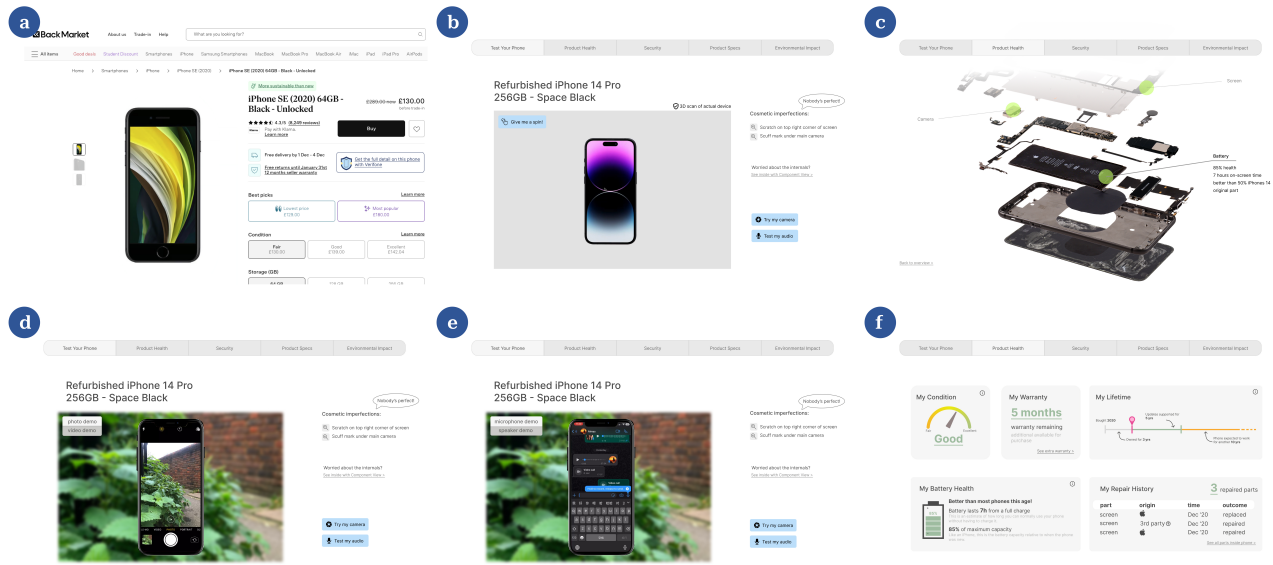


Figure 5: (a) Product listing on BackMarket [6], with Verifone icon added; (b) 3D interactive phone model; (c) Interactive exploded model; (d) Camera testing feature; (e) Audio testing feature; (f) Product Health page.

device, an important factor given the individuality of each second-hand device. A participant noted the affective prototype seemed to be “created by real people... giving me some sort of personality... so there is a bit more trust there”.

In response to feedback, we refined our designs, adopting a hybrid approach that blends the affective prototype’s personality with the functional prototype’s clarity. Using the functional prototype’s typeface and structure for a clean aesthetic, we adjusted the UX copy to convey a more informal tone, updated headings to include ‘My’ to underscore device-specific data, and retained humour with the landing page comment, ‘Nobody’s perfect!’. Based on user feedback, tooltips were added to clarify data collection and calculation.

7 FINAL DESIGN

Our final design of Verifone acts as a third-party verifier in the purchasing journey that stores and presents data for registered phones identified by unique serial numbers. It meets our user requirements in the following ways:

7.1 Accessible: Available on seller listings

Users see a widget that opens a new tab to Verifone (fig.5a) from within the seller listing. This fits into the existing purchasing journey and doesn’t require users to search on another platform.

7.2 Reliable: Opportunity to inspect the phone

On the landing page, Verifone presents a 3D interactive model of the specific phone for users to visually inspect its external condition (fig.5b). Using the actual phone instead of a stock model enables users to identify specific cosmetic imperfections, ensuring a more

accurate assessment, especially for phones labelled as being in ‘fair’ condition.

For more technical users, a detailed ‘exploded’ model with hotspots is available at the bottom of the page (fig.5c). This discreet link caters to users seeking detailed information on specific components, as this view can be overwhelming for less technical users based on our evaluations. Users can also assess the phone’s camera and audio quality through video simulations (fig.5d-e). This experience allows users to review photos, videos, and voice notes recorded with the phone, providing a more accurate understanding of the phone’s functionality and enhancing the user’s ability to envision using the device.

7.3 Clear: Breaks down condition of the phone

Verifone’s product health page offers crucial information in simplified data visualisations that help users gauge the phone’s overall condition, warranty, battery life, expected longevity and repair history (fig.5f). This section aids the user in grasping the value of the phone. For example, we present a timeline indicating the purchase date, how long users can expect software updates for, and the phone’s functional lifespan. The repair history section, crucial for future repairs or resale, is presented in tabular format, detailing when specific parts were repaired or replaced and whether they came from the original manufacturer or a third party.

8 DISCUSSION & FUTURE WORK

This paper introduces Verifone, a digital interface that seeks to remedy the lack of trust in the second-hand phone market, rooted in information asymmetry between sellers and buyers. To overcome this challenge, we drew inspiration from Akerlof’s “The Market for

Lemons” to design trust-building measures, incorporating device-specific interactive elements and highlighting key data about the phone’s condition.

Our initial designs included both functional and affective styles. Usability testing showed that the functional style engendered more trust in users overall, perhaps due to increased readability and familiarity from aligning with established design conventions. However, certain affective elements were seen to highlight the device’s uniqueness. This is a departure from the conventional mental model in most digital stores, where the product depicted is typically a sample or abstract representation rather than the specific device users expect to receive.

Our final design combines features from both prototypes, blending the minimal aesthetics of the functional prototype with UX copy conveying a unique sense of personality from the affective one. Our findings suggest that Verifone can act as a reliable source of information when potential buyers browse for second-hand phones, boosting confidence to follow through with the purchase. In the long run, this could lead to increased activity in the used phone market, promote sustainability, reduce electronic waste, and foster a circular economy. Our research suggests context plays a crucial role in determining the effectiveness of “cute” design elements in building trust.

Future work should refine and iterate Verifone through testing with a larger sample size to validate and extend its effectiveness in building trust. This includes addressing identified limitations from usability tests, like the lack of seller and pricing information, and the need for greater emphasis on device-specific details. Additional research is needed to understand sellers and their needs within the used phone market. Exploring strategies to engage and incentivise sellers, repair shops, and manufacturers to provide accurate device information in a shared information uploading system, with potential support from regulatory bodies, would contribute to building a more transparent and trustworthy circular ecosystem.

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